MEDTRACK DV

Hospital Performance Intelligence System for Operational and Patient Care Analytics

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Project Documentation
and Dashboard Presentation

Project Detail	**Description**
Project Domain	Healthcare Analytics / Data Visualization
Primary Tools	Python, Google Colab, Tableau, CSV datasets
Dashboard Sections	Hospital Overview, Patient Flow, Department Analytics, Resource Utilization
Project Type	Healthcare Data Analytics and Visualization

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1. Project Overview

1.1 Introduction

MEDTRACK DV is a hospital analytics and visualization project designed to organize healthcare information into a structured dashboard environment.

The project transforms source healthcare records into analysis-ready datasets and presents important measures and patterns through Tableau worksheets and dashboards.

The dashboard focuses on hospital-wide activity, patient flow, department-level indicators, and resource utilization.

1.2 Problem Statement

Hospital data is often distributed across multiple tables and files. Reviewing admissions, length of stay, readmissions, department activity, and bed utilization separately can make it difficult to understand the overall operational picture.

This project consolidates selected healthcare data into analytical views so users can monitor activity and compare patterns across departments, wards, and time periods.

1.3 Project Objectives

- Prepare and standardize healthcare data for consistent analysis.
- Create analytical datasets for hospital overview, patient flow, department analytics, and resource utilization.
- Calculate and validate hospital operational KPIs.
- Build Tableau worksheets and dashboards for monitoring volume, patient flow, department patterns, and bed usage.
- Present findings in a clear visual format that supports project review and operational discussion.

1.4 Project Scope

Table 1.1: Project Scope

In Scope	**Not Claimed**
Descriptive and comparative analysis of supplied healthcare datasets.	Clinical diagnosis, treatment recommendations, or patient-level medical advice.
KPI reporting and visual exploration in Tableau.	Causal conclusions about why a department or outcome differs.
Patient counts, stay duration, readmission measures, and bed utilization views.	Live hospital-system integration or real-time monitoring unless separately connected.

1.5 Expected Users

The dashboard is intended for project reviewers and healthcare operations stakeholders who need a summarized view of patient volume, service patterns, department-level indicators, and resource utilization.

2. Data Sources and Preparation

2.1 Data Sources

The project used multiple healthcare CSV files rather than a single flat source.

Data preparation was performed using Python in Google Colab, with files organized in Google Drive.

The prepared analytical datasets were then connected to Tableau for visualization.

2.2 Data Preparation Workflow

Stage	**Work Performed**	**Purpose / Output**
Source Collection	Collected source CSV files for patient, admission, bed, billing, department, diagnosis, and related records.	Multi-file healthcare source collection.
Profiling	Reviewed dataset dimensions, columns, missing values, duplicates, and key fields.	Understanding of data quality and available analysis fields.
Cleaning	Removed identified duplicate records and handled missing values according to the cleaning workflow.	Cleaned and more consistent data.
Standardization	Standardized department and identifier-related fields and prepared date fields.	Consistent dimensions for grouping and linking.
Analytical Modeling	Prepared four analytical datasets for Tableau.	Hospital overview, patient flow, department analytics, and resource utilization.

Validation	Checked linking keys and validated key measures against project calculations.	Cross-checks for dashboard metrics.

2.3 Analytical Dataset Inventory

Table 2.2: Prepared Analytical Datasets

Dataset	**Dimensions**	**Analytical Role**
hospital_overview.csv	10,000 × 11	Hospital-level admissions and overview measures.
patient_flow.csv	10,000 × 12	Admission/check-out timing, patient movement, and stay analysis.
department_analytics.csv	9,944 × 10	Department-level volume, satisfaction, and readmission analysis.
resource_utilization.csv	9,838 × 7	Bed availability, occupancy, and resource utilization views.

2.4 Data Quality Notes

The source preparation identified 4,109 duplicate rows in the large inpatient-discharge source and removed them during cleaning.

Missing dates were present in the readmission dataset. Therefore, date-based analysis should be interpreted only over the available records and date range.

3. Key Performance Indicators

3.1 KPI Summary

The following table consolidates the principal KPI values recorded during project validation.

KPI	**Reported Value**	**Meaning / Calculation**
Total Admissions	10,000	Count of admission or patient records in the overview dataset.
Total Beds	2,991,110	Reported total bed count or capacity measure in project KPI calculations.
Occupied Beds	1,990,850	Reported occupied-bed measure used in the occupancy calculation.
Bed Occupancy Rate	66.56%	Occupied beds divided by total beds, expressed as a percentage.
Average Length of Stay	15.51 days	Average recorded length of stay.
Readmission Rate	41.42%	4,142 readmitted records among 10,000 eligible records.
Total Readmissions	4,142	Count of records classified as readmitted in the validated KPI calculation.
Department Efficiency Score	Mean: 52.03 Minimum: 1.29 Maximum: 99.4	Efficiency metric referenced in project calculations. See data limitation below.

3.2 KPI Interpretation

3.2.1 Total Admissions

Total admissions represent the number of patient admission records counted in the overview dataset.

The project used Count of Patient Id for the total-admission KPI.

3.2.2 Bed Occupancy Rate

Bed occupancy rate represents the relationship between occupied beds and the selected total-bed denominator.

$$OccupancyRate = \frac{OccupiedBeds}{TotalBeds} \times 100$$

The reported project value was 66.56%.

3.2.3 Average Length of Stay

Average length of stay represents the average recorded duration of patient stay.

The reported project value was 15.51 days.

3.2.4 Readmission Rate

The project recorded 4,142 readmitted records among 10,000 eligible records.

$$ReadmissionRate = \frac{ReadmittedRecords}{EligibleRecords} \times 100$$

The calculated project value was 41.42%.

3.3 KPI Validation Notes

- Counts depend on the chosen record grain and aggregation used in Tableau.
- A second bed-utilization calculation in project notes reported 66.67%, based on 1,969,490 occupied and 2,953,930 available.
- Readmission rate should be described with the project-defined eligible population and time window.

- The department_efficiency_score field was noted as absent from the delivered department_analytics.csv. The reported efficiency summary should not be presented as a directly reproducible Tableau field until the calculation is added and validated.

3.4 KPI Presentation

Each KPI is intended to be displayed as a labeled tile with the metric name above the prominent numerical value.

The metric definition or date period can be included in a tooltip or dashboard subtitle.

4. Hospital Overview Dashboard

4.1 Dashboard Purpose

The Hospital Overview dashboard serves as the landing dashboard of MEDTRACK DV.

It summarizes admission volume, disease mix, monthly activity, age distribution, and selected headline KPIs.

4.2 Worksheets and Charts

Worksheet / Visual	**Chart Type**	**Analytical Purpose**
KPI Admissions	KPI Text Card	Displays total admission count (10,000 in project validation).
KPI Occupancy	KPI Text Card	Displays the occupancy percentage using the selected project definition.
Average Length of Stay	KPI Text Card	Displays average stay duration (15.51 days).
Readmission Rate	KPI Text Card	Displays the readmission rate (41.42%).
Disease vs. Patients	Ranked Bar Chart	Compares patient count across disease or diagnosis categories.
Monthly Operational Trends	Line Chart	Shows admission volume over time using admission date at month level.
Discharge Status	Categorical Chart	Summarizes available discharge-status information.
Age vs. Patients	Pie Chart	Shows patient distribution across age groups.

4.3 Dashboard Interpretation

KPI tiles provide the overall snapshot, while disease and age charts help inspect patient composition.

The monthly trend can be used to identify changes in recorded admissions across the available period.

The largest disease or age category represents the largest recorded group in this dataset. It should not be interpreted as population-level disease prevalence.

4.4 Design Choices

KPI worksheets use text marks with a descriptive label and prominent value.

The age pie chart uses color to distinguish age bands, with labels positioned inside slices when space allows.

For category comparisons with many values, a sorted bar chart is more precise than a pie chart.

5. Patient Flow Dashboard

5.1 Dashboard Purpose

The Patient Flow dashboard focuses on the movement of patient records through admission, check-out, ward assignment, and length-of-stay analysis.

5.2 Worksheets and Charts

Worksheet	**Chart Type**	**Purpose and Field Logic**
Admission Trends	Line Chart	Counts Patient Id by Admission Date at month level.
Discharge Tracking	Line / Time Chart	Uses Patient Check Out Date as the available discharge proxy.
Patient Movement by Ward	Treemap	Uses Ward as category and patient count as size to show distribution across wards.
Average Stay Analysis	Bar Chart	Compares average length of stay across available groups such as ward.
Patients per Race	Categorical Chart	Displays patient counts across race categories where included.
Peak Patient Load Monitoring	Line Chart	Tracks patient count by admission date to identify high-volume periods.

5.3 Date Limitation

The available Patient Check Out Date values were reported to cover June through July only.

The discharge view therefore represents the recorded June–July window, not a complete annual discharge trend.

Admission Date and Patient Check Out Date are different events and should not be substituted for one another.

5.4 Patient Movement Interpretation

The ward treemap shows patient distribution by ward.

It does not establish individual transfer paths or movement sequences because a dedicated transfer or movement event field was not identified in the available fields.

The chart should be titled and described as distribution by ward unless transfer data is added.

5.5 Peak Patient Load

Peak Patient Load Monitoring uses admission date and patient count to identify periods with higher recorded admission volume.

A peak in admission counts is not necessarily the same as simultaneous occupied beds. The latter requires a date-specific census or occupancy measure.

6. Department Analytics Dashboard

6.1 Dashboard Purpose

Department Analytics compares activity and selected service indicators across departments.

The project contains worksheets for department satisfaction, patient volume, readmission, and length of stay.

6.2 Worksheets and Charts

Worksheet	**Chart Type**	**What It Communicates**
Average Patient Satisfaction by Department	Bar Chart	Compares average satisfaction by department.
Patient Volume by Department	Bar Chart	Counts patient records by department and supports ranking by volume.
Readmission by Department	Bar Chart / Rate View	Compares readmission indicator by department. Interpretation depends on the source field definition.
Monthly Length of Stay	Line / Time-Series View	Displays length-of-stay behavior over the available time grain.
Department Efficiency Comparison	Planned KPI / Comparison	The efficiency score was referenced in project notes but was not present in the delivered department dataset.
Treatment Capacity Analysis	Planned Operational View	Should use verified capacity fields if present in the connected data.

6.3 Combined Bar and Line Analysis

A dual-axis chart can combine department patient volume as bars and readmission rate as a line.

This is appropriate when the readmission field is numeric, such as a 0/1 flag averaged to a rate, or is already a validated rate.

The two axes should remain independently scaled unless their units are compatible.

A patient count axis should not be synchronized with a percentage axis.

6.4 Interpretation Cautions

- Average satisfaction is a department-level mean, not the number of satisfied patients.
- Readmission comparisons can be affected by department volume, eligibility definitions, and missing data.
- A satisfaction chart should not be labeled as a comprehensive department efficiency score.

7. Resource Utilization Dashboard

7.1 Dashboard Purpose

The Resource Utilization section focuses on bed-related operational measures and peak-load context.

The workbook includes peak patient load monitoring, average bed utilization rate per department, and bed occupancy rate.

7.2 Worksheets and Charts

Table 7.1: Resource Utilization Worksheets

Worksheet	**Chart Type**	**Purpose**
Peak Patient Load Monitoring	Line Chart	Tracks patient count by admission date and identifies high-volume periods.
Average Bed Utilization Rate per Department	Bar Chart	Compares average reported bed-utilization measure by department.
Bed Occupancy Rate	KPI / Rate Chart	Shows occupied capacity relative to the selected bed denominator.

7.3 Resource Utilization Interpretation

Resource utilization views help summarize the relationship between patient activity and reported bed capacity.

The occupancy percentage depends on the definition of occupied beds and the denominator used in the calculation.

8. Conclusion and Future Scope

8.1 Project Outcomes

The MEDTRACK DV project demonstrates how healthcare records can be prepared and organized into operational dashboards.

The project outcomes include:

- Created a multi-dataset preparation workflow and four analysis-ready CSV datasets.
- Built Tableau views across hospital overview, patient flow, department analytics, and resource utilization.
- Calculated and documented headline measures for admissions, length of stay, readmission, and bed occupancy/utilization.
- Developed a reusable dashboard structure that can be extended as more complete operational fields become available.

8.2 Known Limitations and Recommended Actions

Limitation	**Recommended Action**
Check-out dates are limited to June–July in the available data.	Show the date range clearly and obtain broader discharge-date coverage before annual comparisons.
Ward field supports distribution, not transfer-path analysis.	Add patient transfer events with patient ID, from-ward, to-ward, and timestamp if true movement tracking is required.
Department efficiency score is not included in department_analytics.csv.	Add and validate the score calculation in the analytical pipeline before presenting it as a Tableau measure.

Bed KPI variants use different denominators and reported values.	Choose one documented definition, reconcile source fields, and apply it consistently across cards and charts.
Dashboard metrics depend on dataset grain and missing-value handling.	Add metric definitions, filters, date coverage, and source notes to the final dashboard and report.

8.3 Future Scope

The project can be extended by:

- Incorporating complete admission and discharge date coverage.
- Adding explicit patient transfer events to support movement-path analysis.
- Adding and validating department efficiency and treatment-capacity measures.
- Reconciling bed occupancy definitions across source datasets.
- Adding dashboard filters, tooltips, and documented metric definitions.

8.4 Conclusion

MEDTRACK DV demonstrates how healthcare records can be prepared and organized into operational dashboards.

It provides descriptive visibility into patient volume, admission patterns, department-level comparisons, length of stay, readmission, and bed utilization.

Its results should be interpreted within the coverage and definitions of the supplied datasets.

Extending date coverage, documenting KPI formulas, and adding explicit transfer and capacity fields would improve analytical completeness.

End of Project Documentation